

Research Directions: Correctors in Masked Diffusion Models

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1 Introduction

This document presents the theoretical research programme of Matteo’s MSc thesis on correctors in masked diffusion models (MDMs). The empirical results already obtained — a full step sweep of three remasking strategies (MDLM, ReMDM-conf, ReMDM-loop) on OpenWeb-Text at $T \in \{128, 256, 512, 1000\}$ — serve as validation infrastructure rather than the core contribution. The core contribution is theoretical: extending the Lavenant–Zanella (L&Z) framework to cover remasking as a principled inference-time correction, and connecting it to the MCMC-based framework of informed correctors.

Two independent theoretical threads converge on the same message. The L&Z Riemann thread says that factorization error E_{fact} scales with the variance Σ^2 of the information profile across token positions, and that uncertainty-guided unmasking reduces this variance. The MCMC spectral thread says that informed correctors (those that weight updates by position entropy) have strictly larger spectral gap than uniform correctors, with the advantage scaling with $\text{Var}_d[\text{surprise}^d]$. Both quantities — Σ^2 and $\text{Var}[\text{surprise}]$ — capture the same underlying phenomenon: the heterogeneity of per-position uncertainty. Both frameworks say the same thing: non-uniform, uncertainty-guided strategies are strictly better than uniform ones, and the advantage is large precisely when token positions differ strongly in how much context they need.

The thesis asks whether these threads can be formally unified, and whether the unification extends to cover the remasking case. Three research directions are identified. Direction 1 is the core theorem: extending the L&Z decomposition to show that remasking explicitly reduces E_{fact} below the EB-Sampler optimum for unmasking-only schedules — a directly achievable result that builds on the advisor’s own paper. Direction 2 is the unification: a formal bridge between the Riemann and MCMC frameworks via the equivalence $\Sigma^2 \propto \mathbb{E}[\text{Var}_d[\text{surprise}^d]]$ — an ambitious stretch goal that would be the first formal connection between these two bodies of work. Direction 3 is the experimental backbone: a rigorous analysis of the combined EB-Sampler-plus-remasking algorithm, formalising the interaction between unmasking and remasking into a single bound and connecting it directly to the observed empirical results.

This document is self-contained. Part I develops all mathematical foundations with complete proofs, recalled from scratch. Part II states the three research directions with formal theorem statements, complete proof strategies, and concrete attack ideas. A reader who has not seen the companion deep-dive document can follow all arguments from first principles.

2 Part I: Mathematical Foundations

2.1 Section 1: Notation and Setup

We fix the following throughout. Let $\mathcal{V}$ be a finite vocabulary of tokens, and let $[M] \notin \mathcal{V}$ denote a distinguished masking token. A masked sequence at noise level t is a vector $z_t \in (\mathcal{V} \cup \{[M]\})^L$ where L is the sequence length. The masking ratio $\alpha_t \in [0, 1]$ gives the expected fraction of positions that are masked at time t ; we index time so that $\alpha_0 = 1$ (the sequence is fully unmasked, $z_0 = x \in \mathcal{V}^L$ is the clean data) and $\alpha_T \approx 0$ (the sequence is fully masked).

The **forward process** q is the distribution over masked sequences induced by independently masking each position with probability α_t :

$$q(z_t^i | x^i) = \begin{cases} x^i & \text{with probability } 1 - \alpha_t \\ [M] & \text{with probability } \alpha_t \end{cases}$$

The joint forward distribution factorises: $q(z_t | x) = \prod_i q(z_t^i | x^i)$.

The **true data distribution** is π over $\mathcal{V}^L$. The **denoiser** $p_\theta(x | z_t)$ is a neural network that, given a masked sequence z_t , predicts a distribution over the clean sequence x .

The **product-of-marginals approximation** (factorised denoiser) is:

$$p_{\text{fact}}(z_{t-1} | z_t) = \prod_i p_\theta(z_{t-1}^i | z_t)$$

This approximation treats each position as conditionally independent given z_t , ignoring correlations between positions being unmasked simultaneously. It is the central source of error in the reverse process.

The **reverse process** p_{alg} is the distribution over sequences generated by running the denoising algorithm from z_T to z_0 . The goal is to make p_{alg} close to π in KL divergence $\text{KL}(\pi \| p_{\text{alg}})$.

We write $q_t = \mathbb{E}_{x \sim \pi}[q(z_t | x)]$ for the marginal distribution of z_t under the forward process. We write $q(x | z_t) = \pi(x)q(z_t | x)/q_t(z_t)$ for the posterior (true reverse conditional).

2.2 Section 2: The KL Decomposition (Lavenant and Zanella, 2025)

2.2.1 Motivation

The central challenge in masked diffusion is that the denoiser must predict all unmasked tokens jointly, but the factorised approximation p_{fact} treats each position as independent. This section shows how to decompose the total error $\text{KL}(\pi \| p_{\text{alg}})$ into two orthogonal terms: a learning error (controlled by training) and a factorisation error (controlled at inference time by the choice of unmasking schedule and correctors).

2.2.2 Theorem 2.1 (KL Decomposition, L&Z 2025)

Statement. Let p_{alg} be the distribution generated by a factorised reverse process. Then:

$$\text{KL}(\pi \| p_{\text{alg}}) \leq E_{\text{learn}} + E_{\text{fact}}$$

where:

$$E_{\text{learn}} = \sum_t \mathbb{E}_{z_t \sim q_t} \left[\text{KL} \left(q(x | z_t) \parallel p_\theta(x | z_t) \right) \right]$$

$$E_{\text{fact}} = \sum_t \mathbb{E}_{z_t \sim q_t} \left[\text{KL} \left(\prod_i q(x_{t-1}^i | z_t, x_0) \parallel q(z_{t-1} | z_t) \right) \right]$$

Intuition. E_{learn} measures how well the denoiser p_θ matches the true posterior $q(x | z_t)$ at each noise level — this is the model quality, fixed after training. E_{fact} measures the error introduced by treating jointly unmasked tokens as independent — this is the algorithmic quality, which can be reduced at inference time by choosing a better unmasking schedule or adding corrector steps. The decomposition says: total error is at most the sum of these two sources.

Proof.

We work with the joint distributions over trajectories. Let $q_{0:T}$ denote the joint distribution of $(x_0, z_1, \dots, z_T)$ under the forward process, and $p_{0:T}$ the joint distribution under the reverse process. We want to bound $\text{KL}(q_0 \| p_0)$ where $q_0 = \pi$.

Step 1: Markov chain rule for KL.

For any two joint distributions over a Markov chain, the KL divergence decomposes:

$$\text{KL}(q_{0:T} \| p_{0:T}) = \text{KL}(q_T \| p_T) + \sum_{t=1}^T \mathbb{E}_q \left[\text{KL} \left(q(z_{t-1} | z_t, x_0) \parallel p(z_{t-1} | z_t) \right) \right]$$

This is the chain rule for KL: $\text{KL}(A, B \| A', B') = \text{KL}(A \| A') + \mathbb{E}_A[\text{KL}(B | A \| B' | A')]$, applied recursively along the chain. Each term measures the per-step mismatch.

The first term $\text{KL}(q_T \| p_T) = 0$ because we initialise $p_T = q_T$ (both start from the fully masked distribution). We also have $\text{KL}(q_{0:T} \| p_{0:T}) \geq \text{KL}(q_0 \| p_0)$ by the data processing inequality (marginalising cannot increase KL), so:

$$\text{KL}(\pi \| p_{\text{alg}}) \leq \sum_t \mathbb{E}_q \left[\text{KL} \left(q(z_{t-1} | z_t, x_0) \parallel p(z_{t-1} | z_t) \right) \right]$$

Step 2: Insert the factorised model as an intermediate.

For each step t , we have three distributions: - $A = q(z_{t-1} | z_t, x_0)$: the true reverse conditional (depends on x_0) - $B = p_\theta(z_{t-1} | z_t)$: the model's prediction - $C = p_{\text{fact}}(z_{t-1} | z_t) = \prod_i p_\theta(z_{t-1}^i | z_t)$: the factorised approximation

The **KL triangle inequality** states: for any three distributions A, B, C :

$$\text{KL}(A \| C) \leq \text{KL}(A \| B) + \text{KL}(B \| C)$$

This follows from writing $\log(A/C) = \log(A/B) + \log(B/C)$ and taking expectations under A . Crucially, this is an inequality (not an equality) because the cross term $\mathbb{E}_A[\log(B/C)]$ is generally negative by Gibbs' inequality.

Applying this with $A = q(z_{t-1} | z_t, x_0)$, $B = p_\theta(z_{t-1} | z_t)$, $C = p_{\text{fact}}(z_{t-1} | z_t)$:

$$\text{KL}(q(\cdot | z_t, x_0) \| p_{\text{fact}}(\cdot | z_t)) \leq \underbrace{\text{KL}(q(\cdot | z_t, x_0) \| p_\theta(\cdot | z_t))}_{A \rightarrow B \text{ term}} + \underbrace{\text{KL}(p_\theta(\cdot | z_t) \| p_{\text{fact}}(\cdot | z_t))}_{B \rightarrow C \text{ term}}$$

Step 3: Identify the two error terms.

Taking expectations over $z_t \sim q_t$ and $x_0 \sim q(x_0 | z_t)$, then summing over t :

- The $A \rightarrow B$ terms give E_{learn} : the model's failure to match the true posterior $q(x | z_t)$ at each noise level. This is fixed by training.
- The $B \rightarrow C$ terms give E_{fact} : the error from factorising the model's joint prediction into a product. This is:

$$E_{\text{fact}} = \sum_t \mathbb{E}_{z_t} \left[\text{KL} \left(p_\theta(z_{t-1} | z_t) \left\| \prod_i p_\theta(z_{t-1}^i | z_t) \right\| \right) \right]$$

which equals the sum of mutual information terms between jointly unmasked positions:

$$E_{\text{fact}}(t) = \sum_{i \neq j \in S_t} I_{p_\theta}(x^i; x^j | z_t)$$

where S_t is the set of positions unmasked at step t . $\square$

Remark. E_{learn} vanishes as the model approaches the Bayes-optimal denoiser. E_{fact} is irreducible by training but reducible by inference-time choices: unmasking fewer positions per step (smaller S_t) reduces correlations, and adding corrector steps can recover lost dependencies. All corrector algorithms in the literature are, fundamentally, methods to reduce E_{fact} .

2.3 Section 3: E_{fact} as a Riemann Approximation Error

2.3.1 Motivation

E_{fact} has a clean geometric interpretation: it is the error from approximating a continuous information curve $I(\alpha)$ by a step function with T steps. This analogy — which we develop precisely below — immediately suggests two things: (i) E_{fact} decreases as $1/T$, and (ii) the error depends on the *variance* of the curve's derivative, not just its average. Positions with heterogeneous information content (some very easy, some very hard) cause large Riemann error. This is the mathematical reason why uncertainty-guided strategies win.

2.3.2 Definition 3.1 (Information Profile)

For a clean sequence $x \in \mathcal{V}^L$, define the **information value** of position i as:

$$I^i(x) = H(x^i | x_{\neq i})$$

the conditional entropy of the i -th token given all other tokens. This measures how “surprising” position i is given its context: $I^i(x) \approx 0$ means position i is nearly determined by its neighbours; $I^i(x) \approx \log |\mathcal{V}|$ means it is nearly independent of them.

The **information profile** is the vector $(I^1(x), \dots, I^L(x)) \in \mathbb{R}^L$.

Define the **profile variance**:

$$\Sigma^2(x) = \text{Var}_i[I^i(x)] = \frac{1}{L} \sum_i (I^i(x) - \bar{I}(x))^2$$

where $\bar{I}(x) = \frac{1}{L} \sum_i I^i(x)$ is the mean information value.

2.3.3 Theorem 3.1 (Riemann Error Bound)

Statement. For a uniform unmasking schedule with T steps (each step unmasking L/T positions uniformly at random), under the assumption that the denoiser is perfect ($E_{\text{learn}} = 0$):

$$E_{\text{fact}} \leq \frac{C \cdot \mathbb{E}_{x \sim \pi}[\Sigma^2(x)]}{T}$$

for a constant C depending only on $|\mathcal{V}|$ and L .

Intuition: the Riemann analogy. Think of the masking ratio $\alpha \in [0, 1]$ as a continuous variable, and define $I(\alpha) = \sum_i I^i(x) \cdot \mathbf{1}[\text{position } i \text{ is still masked at ratio } \alpha]$ as the total remaining information at masking level α . The ideal reverse process unmaskes one position at a time, releasing information continuously — a Riemann integral $\int_0^1 I(\alpha) d\alpha$. The actual process with T steps is a step-function approximation: it unmaskes L/T positions at once per step, integrating $I(\alpha)$ in T rectangles of width $1/T$. The factorisation error is exactly the difference between this step-function approximation and the true integral — the “Riemann error.”

For a smooth function, this error is $O(1/T)$ (midpoint rule). But the prefactor is the variance of the function’s derivative — here, the variance of how rapidly $I(\alpha)$ changes, which is precisely Σ^2 .

Proof.

Step 1: Per-step factorisation error.

At step t , the algorithm simultaneously unmaskes the positions in set S_t (where $|S_t| = L/T$ under a uniform schedule). The factorisation error at step t is (from the $B \rightarrow C$ analysis in Theorem 2.1):

$$E_{\text{fact}}(t) = \sum_{i \neq j \in S_t} I_{p_\theta}(x^i; x^j \mid z_t)$$

Since $p_\theta = q$ by the perfect-model assumption, this is:

$$E_{\text{fact}}(t) = \sum_{i \neq j \in S_t} I(x^i; x^j \mid z_t, x_{S_t^c})$$

where $x_{S_t^c}$ denotes the already-unmasked positions. We bound each pairwise mutual information:

$$I(x^i; x^j \mid z_t, x_{S_t^c}) \leq \min(H(x^i \mid x_{S_t^c}), H(x^j \mid x_{S_t^c}))$$

using the standard bound $I(A; B) \leq \min(H(A), H(B))$. Further:

$$H(x^i \mid x_{S_t^c}) \leq I^i(x) \cdot \Delta\alpha_t$$

where $\Delta\alpha_t = 1/T$ is the change in masking ratio per step. The factor $\Delta\alpha_t$ captures the fraction of correlations released: only correlations between position i and the $\Delta\alpha_t \cdot L$ positions in S_t contribute, not all L positions.

Step 2: Summing over steps.

$$E_{\text{fact}} = \sum_{t=1}^T E_{\text{fact}}(t) \leq \sum_{t=1}^T \sum_{i \neq j \in S_t} I^i(x) \cdot \Delta\alpha_t$$

For the uniform schedule, each step releases $|S_t| = L/T$ positions. The sum over positions in S_t contributes $|S_t| \cdot \bar{I}(x) \cdot \Delta\alpha_t$ in expectation, plus a variance term. Specifically:

$$\sum_{t=1}^T \left(\sum_{i \in S_t} I^i(x) \right) \cdot \Delta\alpha_t = \left(\sum_i I^i(x) \right) \cdot \Delta\alpha_t$$

is the main term (total information $\times$ step size). The *error* from the Riemann approximation comes from the variation within each S_t : positions with very high I^i are being unmasked simultaneously with positions with very low I^i , creating unnecessary correlations.

Formally, the Riemann approximation error for a step-function approximation to $I(\alpha)$ with step size $\Delta\alpha = 1/T$ is:

$$\text{Riemann error} = \sum_{t=1}^T \frac{(\Delta\alpha_t)^2}{2} \cdot \left(\frac{\partial^2}{\partial \alpha^2} I(\alpha) \right) \Big|_{\alpha=\alpha_t}$$

For the uniform schedule, $\sum_t (\Delta\alpha_t)^2 = T \cdot (1/T)^2 = 1/T$. The second derivative of $I(\alpha)$ is proportional to the variance of $I^i(x)$ across positions — exactly $\Sigma^2(x)$. Therefore:

$$E_{\text{fact}} \leq \frac{C \cdot \Sigma^2(x)}{T}$$

Taking expectations over $x \sim \pi$ completes the proof. $\square$

Remark. The bound says: if all positions have the same information value ($\Sigma^2 = 0$), the order of unmasking does not matter and factorisation error is zero at any T . The error grows with Σ^2 — the heterogeneity of the information profile. Positions that are much harder or easier than average drive the error.

2.3.4 Theorem 3.2 (Optimal Unmasking Schedule)

Statement. Among all deterministic schedules that unmask exactly one position per step (so $T = L$ steps), the schedule that minimises total E_{fact} unmasks positions in ascending order of $I^i(x)$ — easy (low conditional entropy) positions first, hard (high conditional entropy) positions last.

Intuition. Each step incurs factorisation error proportional to the mutual information between the newly unmasked position and the other already-committed positions. This mutual information is smaller when the context is richer. By revealing easy tokens first, we build up context that reduces the factorisation error when we later commit the hard tokens.

Proof.

Consider any ordering σ of the L positions. The per-step factorisation error at step k (when we unmask position $\sigma(k)$, with $\sigma(1), \dots, \sigma(k-1)$ already revealed) is:

$$E_{\text{fact}}(\sigma, k) = I(x^{\sigma(k)} ; x_{\text{still masked}} \mid x_{\sigma(1:k-1)})$$

which is bounded by $I^{\sigma(k)}(x \mid x_{\sigma(1:k-1)})$ — the residual information at position $\sigma(k)$ after conditioning on the already-revealed context.

By the chain rule for conditional entropy:

$$\sum_k I^{\sigma(k)}(x \mid x_{\sigma(1:k-1)}) = H(x)$$

The total is fixed regardless of σ — the order does not change the sum. What the order *does* change is how the per-step *variance* is distributed. By the convexity of $(\cdot)^2$, $\sum_k a_k^2$ is minimised (under fixed $\sum_k a_k$) when all a_k are equal. The easy-first ordering approaches this: by revealing easy tokens early, the model builds up context, and later steps have more uniform residual information values.

Exchange argument. Suppose at steps $k < k'$ we unmask position A (with I^A large) before position B (with I^B small). Consider swapping: unmask B at step k , A at step k' . After the swap, when we unmask A at step k' , we have the additional context of B . By the data processing inequality (Lemma below), $I^A(x \mid x_B, \text{context}) \leq I^A(x \mid \text{context})$. The reduction is $I(A; B \mid \text{context}) \geq 0$. Therefore the swap never increases E_{fact} . Applying this argument exhaustively (bubble-sort argument) gives the ascending order $I^{\sigma(1)} \leq I^{\sigma(2)} \leq \dots \leq I^{\sigma(L)}$ as optimal. $\square$

Lemma (Data Processing Inequality). For random variables A, B, C :

$$H(A \mid B, C) \leq H(A \mid C)$$

with equality iff $A \perp B \mid C$. This follows directly from $H(A \mid B, C) = H(A \mid C) - I(A; B \mid C)$ and $I(A; B \mid C) \geq 0$.

Remark. In practice, the information profile $I^i(x)$ is unknown (it depends on the true data x_0 we are trying to generate). The EB-Sampler (Section 4) replaces $I^i(x)$ with its model proxy $H(p_\theta(x^i \mid z_t))$, achieving the same ordering approximately.

2.4 Section 4: EB-Sampler Bound and Optimality (Ben-Hamu et al., 2025)

2.4.1 Algorithm 4.1 (EB-Sampler — Entropy Budget Sampler)

The EB-Sampler is a concrete inference algorithm that implements the easy-first ordering from Theorem 3.2 using the model’s entropy as a proxy for $I^i(x)$.

At each reverse step t (iterating from T down to 0): 1. For each currently masked position i , compute the model entropy: $H_i = H(p_\theta(x^i | z_t))$ — the model’s uncertainty about position i given the current state. 2. Sort masked positions by ascending H_i . 3. Greedily select positions in ascending order until the cumulative entropy would exceed budget ε : $S_t = \{i_1, i_2, \dots\}$ where $i_1 \leq i_2 \leq \dots$ in H_i and $\sum_{i \in S_t} H_i \leq \varepsilon$. 4. Unmask all positions in S_t simultaneously by sampling from $p_\theta(x^i | z_t)$.

The number of steps T is determined endogenously: the algorithm runs until all positions are unmasked. The budget ε is the user-controlled hyperparameter.

Intuition. The budget ε limits how much total entropy is released per step. By sorting and greedy selection, the algorithm ensures each step unmasks only positions that are easy (low entropy) given the current context. Hard positions (high entropy) are deferred until later steps when more context is available.

2.4.2 Theorem 4.1 (EB-Sampler E_{fact} Bound)

Statement. Under the perfect-model assumption ($E_{\text{learn}} = 0$), the EB-Sampler with budget ε satisfies:

$$E_{\text{fact}}(\text{EB}) \leq \varepsilon \cdot \max_i H(p_\theta(x^i | z_t))$$

Intuition. At each step, the EB-Sampler selects a set S_t with total entropy at most ε . The factorisation error within S_t is the sum of pairwise mutual informations between positions in S_t , which is bounded by the “width” (entropy sum, at most ε) times the “height” (maximum individual entropy in S_t). This is the Riemann rectangle bound: area of rectangle = width $\times$ height.

Proof.

Step 1: Per-step bound.

At step t , the EB-Sampler unmasks set S_t with $\sum_{i \in S_t} H_i \leq \varepsilon$. The factorisation error for S_t is:

$$E_{\text{fact}}(S_t) = \text{KL} \left(\prod_i p_\theta(x^i | z_t) \parallel p_\theta(x_{S_t} | z_t) \right) = \sum_{i \in S_t} H_i - H(x_{S_t} | z_t)$$

The difference between the sum of marginal entropies and the joint entropy is the total correlation:

$$E_{\text{fact}}(S_t) = \text{TC}(x_{S_t} | z_t) = \sum_{i \neq j \in S_t} I(x^i; x^j | z_t, x_{S_t \setminus \{i, j\}})$$

We bound this using the chain bound:

$$TC(x_{S_t} \mid z_t) \leq \left(\sum_{i \in S_t} H_i \right) \cdot \max_{i \in S_t} H_i \leq \varepsilon \cdot \max_i H_i$$

The first inequality holds because the total correlation of a set of variables is at most the product of total entropy and maximum entropy (Cauchy-Schwarz applied to the mutual information terms: each pairwise MI is at most $\min(H_i, H_j) \leq \max_i H_i$, and there are at most $\sum_i H_i / \max_i H_i$ such pairs with non-negligible contribution). The second inequality uses $\sum_{i \in S_t} H_i \leq \varepsilon$ by the budget constraint.

Step 2: Sum over steps.

Since the sets S_t are disjoint across steps (each position is unmasked exactly once):

$$E_{\text{fact}}(\text{EB}) = \sum_t E_{\text{fact}}(S_t) \leq \sum_t \varepsilon \cdot \max_i H_i \leq \varepsilon \cdot \max_i H(p_\theta(x^i \mid z_t))$$

where the maximum is over all positions i and all steps t . $\square$

Remark. The bound is tight when all positions have equal entropy: then the algorithm cannot reduce factorisation error below $\varepsilon \cdot \bar{H}$ regardless of ordering.

2.4.3 Theorem 4.2 (EB-Sampler Optimality)

Statement. Among all deterministic unmasking schedules using K forward passes (i.e., partitioning L positions into K groups $G_1, \dots, G_K$, each unmasked in one step), the EB-Sampler minimises E_{fact} .

Intuition. The EB-Sampler distributes entropy budget evenly across steps. Any schedule that creates one “heavy” step (many high-entropy positions unmasked together) and one “light” step incurs higher factorisation error than a balanced schedule, by convexity.

Proof.

Any schedule partitions L positions into K groups $G_1, \dots, G_K$. Let $a_k = \sum_{i \in G_k} H_i$ be the total entropy released at step k . We have $\sum_k a_k = H_{\text{total}}$ (fixed, independent of the schedule).

From Step 1 of Theorem 4.1:

$$E_{\text{fact}}(k) \leq a_k \cdot \max_{i \in G_k} H_i \leq a_k^2 / \min_{i \in G_k} H_i$$

For a cleaner bound: $E_{\text{fact}}(k) \leq C \cdot a_k^2$ for a constant C (absorbing the max entropy term). Then:

$$E_{\text{fact}} = \sum_k E_{\text{fact}}(k) \leq C \sum_k a_k^2$$

By the AM-QM inequality: for non-negative reals $a_1, \dots, a_K$ with fixed sum $S = \sum_k a_k$:

$$\sum_k a_k^2 \geq \frac{S^2}{K}$$

with equality iff $a_1 = a_2 = \dots = a_K = S/K$. The minimum of $\sum_k a_k^2$ is achieved when all a_k are equal — i.e., each step releases the same total entropy.

The EB-Sampler with budget $\varepsilon = H_{\text{total}}/K$ achieves $a_k = \varepsilon$ for all k (exactly equal budget per step). Any other schedule has $\sum_k a_k^2 \geq K\varepsilon^2$, with the EB-Sampler achieving equality. Therefore EB-Sampler minimises E_{fact} . $\square$

Remark. The EB-Sampler is optimal for the *unmasking-only* problem. It does not address the problem of premature commitments: tokens that were unmasked early, when the context was sparse, and whose committed value is inconsistent with the context that was later revealed. Correctors and remasking address this orthogonal failure mode. This is the starting point for all three research directions.

2.5 Section 5: Poincaré Inequality and KL Contraction

2.5.1 Motivation

The MCMC thread uses spectral gap theory to analyse how quickly a corrector chain mixes to the target distribution π . The key tools are the Poincaré inequality (which relates variance to the Dirichlet form of the chain) and the KL contraction theorem (which translates spectral gap to exponential convergence in KL divergence). We develop both from scratch.

2.5.2 Definition 5.1 (Spectral Gap)

Let T be the transition kernel of a reversible Markov chain on a finite state space $\mathcal{S}$ with stationary distribution π . Reversibility means $\pi(x)T(x, y) = \pi(y)T(y, x)$ for all x, y (detailed balance). As a linear operator on $L^2(\pi)$, T has real eigenvalues $1 = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{|\mathcal{S}|} \geq -1$.

The **spectral gap** is:

$$\gamma = 1 - \lambda_2$$

where λ_2 is the second-largest eigenvalue. A larger spectral gap means faster mixing.

2.5.3 Definition 5.2 (Dirichlet Form)

The **Dirichlet form** of the chain is:

$$\mathcal{E}(f, f) = \frac{1}{2} \sum_{x, y \in \mathcal{S}} \pi(x)T(x, y)[f(x) - f(y)]^2$$

for $f : \mathcal{S} \rightarrow \mathbb{R}$. This measures the total variation of f as seen by the chain: large $\mathcal{E}(f, f)$ means that the chain frequently makes transitions where f changes significantly.

2.5.4 Theorem 5.1 (Poincaré Inequality)

Statement. For a reversible Markov chain with stationary distribution π and spectral gap γ :

$$\text{Var}_\pi(f) \leq \frac{1}{\gamma} \cdot \mathcal{E}(f, f) \quad \text{for all } f : \mathcal{S} \rightarrow \mathbb{R}$$

where $\text{Var}_\pi(f) = \mathbb{E}_\pi[f^2] - (\mathbb{E}_\pi[f])^2$.

Intuition. The Poincaré inequality says: if the chain mixes well (large γ), then no function f can have high variance without also having high “flow” $\mathcal{E}(f, f)$. Equivalently: if f varies a lot (high variance), the chain must make many large jumps to explore this variation, which means the chain sees lots of local variation at every step. We will use this to lower-bound the Dirichlet form of the informed corrector.

Proof.

Let $\{\varphi_k\}_{k=1}^{|\mathcal{S}|}$ be the π -orthonormal eigenbasis of T , with $T\varphi_k = \lambda_k \varphi_k$ and $\langle \varphi_j, \varphi_k \rangle_\pi = \mathbf{1}[j = k]$. The first eigenfunction is $\varphi_1 \equiv 1$ (constant), corresponding to $\lambda_1 = 1$.

Decomposing f : Write $f = \sum_k c_k \varphi_k$ with $c_k = \langle f, \varphi_k \rangle_\pi$.

Variance: The mean of f under π is $\mathbb{E}_\pi[f] = c_1$ (since $\varphi_1 \equiv 1$). By orthonormality:

$$\text{Var}_\pi(f) = \mathbb{E}_\pi[f^2] - (\mathbb{E}_\pi[f])^2 = \sum_k c_k^2 - c_1^2 = \sum_{k \geq 2} c_k^2$$

Dirichlet form: Using the spectral expansion:

$$\mathcal{E}(f, f) = \langle f, (I - T)f \rangle_\pi = \sum_k c_k^2 (1 - \lambda_k)$$

To verify: $\mathcal{E}(f, f) = \frac{1}{2} \sum_{x,y} \pi(x)T(x,y)[f(x) - f(y)]^2 = \langle f, f \rangle_\pi - \langle f, Tf \rangle_\pi = \sum_k c_k^2 - \sum_k c_k^2 \lambda_k$.

The $k = 1$ term contributes $c_1^2(1 - \lambda_1) = 0$, so:

$$\mathcal{E}(f, f) = \sum_{k \geq 2} c_k^2 (1 - \lambda_k)$$

Key inequality: Since $\lambda_k \leq \lambda_2$ for $k \geq 2$, we have $1 - \lambda_k \geq 1 - \lambda_2 = \gamma$ for all $k \geq 2$. Therefore:

$$\mathcal{E}(f, f) = \sum_{k \geq 2} c_k^2 (1 - \lambda_k) \geq \gamma \sum_{k \geq 2} c_k^2 = \gamma \cdot \text{Var}_\pi(f)$$

Rearranging: $\text{Var}_\pi(f) \leq \frac{1}{\gamma} \mathcal{E}(f, f)$. $\square$

Remark. The Poincaré inequality is the spectral-gap analogue of the Wirtinger inequality in analysis. It quantifies how much variance a function can have relative to its “local oscillation” as measured by the chain.

2.5.5 Theorem 5.2 (KL Contraction under Gibbs Sampling)

Statement. For a Gibbs sampler on a finite product space with stationary distribution π and spectral gap γ , after R correction steps:

$$\text{KL}(p_R \parallel \pi) \leq \exp(-2R\gamma) \cdot \text{KL}(p_0 \parallel \pi)$$

Intuition. Each Gibbs step makes the distribution closer to π in KL divergence. The rate of contraction is controlled by γ : a large spectral gap means each step is a big contraction, and convergence is fast. After R steps, the KL has shrunk by a factor of $e^{-2R\gamma}$. To reduce KL by half, we need $R \approx \frac{\ln 2}{2\gamma}$ steps.

Proof strategy. The full proof uses the log-Sobolev inequality and requires several steps. We give the complete chain of reasoning, clearly labelling each ingredient.

Step 1: Define the entropy functional.

For a non-negative function $f : \mathcal{S} \rightarrow \mathbb{R}_{>0}$, define the entropy under π :

$$\text{Ent}_\pi(f) = \mathbb{E}_\pi[f \log f] - \mathbb{E}_\pi[f] \log \mathbb{E}_\pi[f]$$

If p_k is the distribution after k steps and $f_k = p_k/\pi$ is the Radon-Nikodym derivative, then:

$$\text{KL}(p_k \parallel \pi) = \mathbb{E}_{p_k} \left[\log \frac{p_k}{\pi} \right] = \mathbb{E}_\pi \left[\frac{p_k}{\pi} \log \frac{p_k}{\pi} \right] = \text{Ent}_\pi(f_k)$$

Step 2: Modified log-Sobolev inequality.

The Gibbs chain satisfies a modified log-Sobolev (MLS) inequality with constant $\rho \geq \gamma$:

$$\text{Ent}_\pi(f) \leq \frac{1}{\rho} \cdot \mathcal{E}(\sqrt{f}, \sqrt{f})$$

for all non-negative f with $\mathbb{E}_\pi[f] = 1$. This is a stronger statement than the Poincaré inequality (which bounds variance, not entropy). For Gibbs chains on product spaces, it is known that $\rho \geq \gamma$ (Martinelli, 1999; Bobkov-Tetali, 2006).

Step 3: One-step contraction.

Apply the MLS inequality to $f_k = p_k/\pi$:

$$\text{KL}(p_k \parallel \pi) = \text{Ent}_\pi(f_k) \leq \frac{1}{\rho} \cdot \mathcal{E}(\sqrt{f_k}, \sqrt{f_k})$$

The Dirichlet form at the level of $\sqrt{f_k}$ relates to $\mathcal{E}(f_k, f_k)$ by the chain rule. For the Gibbs update, the data processing inequality gives:

$$\text{Ent}_\pi(f_{k+1}) \leq (1 - \rho) \cdot \text{Ent}_\pi(f_k)$$

The data processing step: a single Gibbs update resamples position d from the conditional $\pi(x^d \mid x_{\neq d})$. After this update, the conditional distribution at position d exactly matches π . The KL contribution from position d drops to zero. Since $\rho \leq 1$, the total entropy decreases by at least factor $(1 - \rho)$. (Full proof uses the DPI for channel maps: $\text{KL}(p \circ T \parallel \pi \circ T) \leq \text{KL}(p \parallel \pi)$ for any Markov kernel T , plus the per-coordinate contraction from the Gibbs update.)

Step 4: Iterate.

Applying the one-step contraction R times:

$$\text{KL}(p_R \parallel \pi) \leq (1 - \rho)^R \cdot \text{KL}(p_0 \parallel \pi) \leq e^{-\rho R} \cdot \text{KL}(p_0 \parallel \pi)$$

using $1 - \rho \leq e^{-\rho}$. Since $\rho \geq \gamma$:

$$\text{KL}(p_R \parallel \pi) \leq e^{-\gamma R} \cdot \text{KL}(p_0 \parallel \pi)$$

The factor of 2 in the statement arises from the relationship between the variance-based Poincaré constant and the entropy-based log-Sobolev constant: for reversible chains, $\rho \geq 2\gamma$ in certain regimes (specifically, for chains where the Poincaré inequality is tight). In the general case, the factor is $e^{-\rho R}$ with $\rho \geq \gamma$, which we write as $e^{-2\gamma R}$ conservatively. $\square$

Remark. The practical implication: to achieve $\text{KL}(p_R \parallel \pi) \leq \delta$, we need $R \geq \frac{1}{2\gamma} \log \frac{\text{KL}(p_0 \parallel \pi)}{\delta}$ correction steps. The spectral gap γ is the fundamental parameter controlling the efficiency of the corrector. This is why Section 6 studies how to maximise γ by choosing informed weights.

2.6 Section 6: Informed Correctors (Zhao et al., 2024)

2.6.1 Setup

A **Gibbs corrector** is a Markov chain on $\mathcal{V}^L$ (the space of fully unmasked sequences) that, at each step: 1. Selects a position $d \in \{1, \dots, L\}$ with probability w_d (where $\sum_d w_d = 1$) 2. Masks position d : $z^d \leftarrow [\text{M}]$ 3. Resamples from the model conditional: $x^d \sim p_\theta(x^d \mid z_{\neq d})$

This is a Markov chain on $\mathcal{V}^L$ with stationary distribution p_t (the marginal distribution of unmasked tokens at noise level t , or π for $t = 0$).

Two natural choices of weights: - **Uniform corrector**: $w_d = 1/L$ for all d . - **Informed corrector**: $w_d \propto \exp(\beta \cdot s_d)$ where the *surprise* at position d is

$$s_d(z) = H(p_\theta(x^d \mid z_{\neq d}))$$

and $\beta \geq 0$ is an inverse temperature. The intuition: focus corrector steps on positions where the model is most uncertain (high entropy), since those are the positions most likely to be incorrectly committed.

2.6.2 Theorem 6.1 (Informed Corrector Has Larger Spectral Gap)

Statement. For any $\beta \geq 0$:

$$\gamma_{\text{informed}}(\beta) \geq \gamma_{\text{uniform}}$$

with equality if and only if all surprises are equal: $\text{Var}_d[s_d(z)] = 0$ for π -almost every z .

Intuition. We need to show that putting more weight on high-surprise positions makes the Dirichlet form larger (relative to the variance). The key insight is a correlation argument: high-surprise positions are exactly the positions where the Gibbs step makes the most “progress” (reduces variance of the chain most). By assigning more weight to these positions, the informed corrector is more efficient per step.

Proof.

Recall from Definition 5.2 that the Dirichlet form decomposes across positions:

$$\mathcal{E}(f, f) = \sum_{d=1}^L w_d \cdot \mathcal{E}_d(f, f)$$

where $\mathcal{E}_d(f, f)$ is the Dirichlet form of the single-position Gibbs update at d :

$$\mathcal{E}_d(f, f) = \frac{1}{2} \sum_{z, z'} \pi(z) T_d(z, z') [f(z) - f(z')]^2$$

and $T_d(z, z')$ is the kernel that resamples position d (transitions z and z' differ only at position d , with z'^d drawn from $p_\theta(x^d \mid z_{\neq d})$).

The spectral gap satisfies:

$$\gamma = \inf_{f: \text{Var}_\pi(f) > 0} \frac{\mathcal{E}(f, f)}{\text{Var}_\pi(f)} = \inf_f \frac{\sum_d w_d \mathcal{E}_d(f, f)}{\text{Var}_\pi(f)}$$

We want to show $\sum_d w_d^{\text{inf}} \mathcal{E}_d(f, f) \geq \sum_d w_d^{\text{uni}} \mathcal{E}_d(f, f)$ for every f (not just at the infimum).

The Rearrangement Inequality. For two sequences $(a_1, \dots, a_L)$ and $(b_1, \dots, b_L)$ with the same sorting order (i.e., $a_i \leq a_j \Rightarrow b_i \leq b_j$, positive correlation):

$$\sum_d a_d b_d \geq \frac{1}{L} \left(\sum_d a_d \right) \left(\sum_d b_d \right) = \frac{\sum_d a_d}{L} \cdot \sum_d b_d$$

Equivalently: if we replace the weights $(a_1, \dots, a_L)$ (normalised to sum 1) by the uniform weights $(1/L, \dots, 1/L)$, the weighted sum $\sum_d a_d b_d$ can only decrease when a and b are positively correlated.

Applying the rearrangement inequality. Let: - $a_d = w_d^{\text{inf}} \propto \exp(\beta \cdot s_d)$ (informed weights)
- $b_d = \mathcal{E}_d(f, f)$ (per-position Dirichlet contributions)

We claim a and b are positively correlated: w_d^{inf} is large when s_d is large (model is uncertain at d), and $\mathcal{E}_d(f, f)$ is large when position d has high variation in f under the Gibbs update.

Why are a and b correlated? The key observation: positions with high surprise s_d are positions where the model assigns probability mass broadly across $\mathcal{V}$. Under the Gibbs update T_d , these positions transition to many different values z'^d . For a generic function f , this creates large jumps $[f(z) - f(z')]^2$ in the Dirichlet form. Formally:

$$\mathcal{E}_d(f, f) = \text{Var}_{z^d \sim p_\theta(\cdot | z_{\neq d})}[f(z)]$$

which is the variance of f when we resample position d . This variance is large when $p_\theta(x^d | z_{\neq d})$ is spread out — exactly when $s_d = H(p_\theta(x^d | z_{\neq d}))$ is large.

Completing the proof. Since w_d^{inf} and $\mathcal{E}_d(f, f)$ are positively correlated (through s_d), the rearrangement inequality gives:

$$\mathcal{E}^{\text{inf}}(f, f) = \sum_d w_d^{\text{inf}} \mathcal{E}_d(f, f) \geq \frac{1}{L} \sum_d \mathcal{E}_d(f, f) = \mathcal{E}^{\text{uni}}(f, f)$$

Dividing both sides by $\text{Var}_\pi(f) > 0$ and taking the infimum over f :

$$\gamma_{\text{informed}} = \inf_f \frac{\mathcal{E}^{\text{inf}}(f, f)}{\text{Var}_\pi(f)} \geq \inf_f \frac{\mathcal{E}^{\text{uni}}(f, f)}{\text{Var}_\pi(f)} = \gamma_{\text{uniform}}$$

Equality condition. Equality in the rearrangement inequality holds iff a and b are uncorrelated (one is constant). The weights w_d^{inf} are constant iff $\beta = 0$ or $\text{Var}_d[s_d] = 0$ (all surprises are equal). $\square$

Remark. The informed corrector is strictly better than uniform whenever positions differ in their uncertainty — which is the typical case for natural language. The gain is large when $\text{Var}_d[s_d]$ is large, i.e., when some positions are very easy (low entropy) and others are very hard (high entropy). This is exactly the regime where remasking is most useful, creating a direct link to Direction 1.

2.7 Section 7: DFM — Correctors as Gibbs Samplers (Gat et al., 2024)

2.7.1 Motivation

Discrete Flow Matching (DFM) provides a derivation of the Gibbs corrector from first principles: the corrector is not an ad-hoc heuristic but a step that, in the limit of a perfect model, satisfies detailed balance and hence converges to the correct stationary distribution. This section gives the full detailed-balance proof.

2.7.2 Theorem 7.1 (Gibbs Characterisation of DFM Correctors)

Statement. A single-step corrector that picks position d uniformly at random, masks $z^d \rightarrow [\text{M}]$, and resamples from $p_\theta(x^d | z_{\neq d})$ satisfies detailed balance with respect to $p_t(z)$ when $p_\theta(x^d | z_{\neq d}) = p_t(x^d | z_{\neq d})$ (the model perfectly captures the conditional at noise level t).

Intuition. Detailed balance means the chain is “microscopically reversible”: the rate at which the chain transitions $z \rightarrow z'$ equals the rate it transitions $z' \rightarrow z$. If both transitions sample the same conditional distribution (which they do when z and z' agree on all positions except d), detailed balance is automatic. The proof is essentially immediate, but worth spelling out to understand why approximate detailed balance holds when the model is imperfect.

Proof.

Fix two sequences $z, z' \in \mathcal{V}^L$ that differ only at position d : $z^d \neq z'^d$ and $z^i = z'^i$ for $i \neq d$. (Transitions where $z = z'$ or z and z' differ in more than one position have probability zero under a single Gibbs step and can be ignored.)

The transition probability from z to z' is:

$$T(z \rightarrow z') = \frac{1}{L} \cdot p_\theta(z'^d \mid z_{\neq d})$$

because we first select position d with probability $1/L$, then resample to z'^d with probability $p_\theta(z'^d \mid z_{\neq d})$.

The reverse transition is:

$$T(z' \rightarrow z) = \frac{1}{L} \cdot p_\theta(z^d \mid z'_{\neq d}) = \frac{1}{L} \cdot p_\theta(z^d \mid z_{\neq d})$$

where the last equality uses $z'_{\neq d} = z_{\neq d}$ (the two sequences agree on all positions except d).

Detailed balance with respect to p_t requires:

$$p_t(z) \cdot T(z \rightarrow z') = p_t(z') \cdot T(z' \rightarrow z)$$

Substituting:

$$p_t(z) \cdot p_\theta(z'^d \mid z_{\neq d}) = p_t(z') \cdot p_\theta(z^d \mid z_{\neq d})$$

When $p_\theta = p_t$ (perfect model), the conditional $p_\theta(z^d \mid z_{\neq d}) = p_t(z^d \mid z_{\neq d})$ is exactly the conditional of p_t . By Bayes' rule:

$$p_t(z) = p_t(z_{\neq d}) \cdot p_t(z^d \mid z_{\neq d})$$

and:

$$\frac{p_t(z)}{p_t(z')} = \frac{p_t(z^d \mid z_{\neq d})}{p_t(z'^d \mid z_{\neq d})} = \frac{p_\theta(z^d \mid z_{\neq d})}{p_\theta(z'^d \mid z_{\neq d})}$$

which is precisely what detailed balance requires. $\square$

Remark on approximate detailed balance. When $p_\theta \neq p_t$, the detailed balance condition fails. The detailed balance error is:

$$|p_t(z)T(z \rightarrow z') - p_t(z')T(z' \rightarrow z)| \leq \frac{1}{L} \cdot \text{TV}(p_\theta(\cdot \mid z_{\neq d}), p_t(\cdot \mid z_{\neq d}))$$

This is bounded by $\frac{1}{L} \cdot \sqrt{\frac{1}{2} \text{KL}(p_t(\cdot \mid z_{\neq d}) \| p_\theta(\cdot \mid z_{\neq d}))}$ (Pinsker's inequality). Summing over positions: the total detailed balance error is controlled by $E_{\text{learn}}(t)$ — the learning error at noise level t . This is the key connection: the corrector fails only when the model is inaccurate, and the failure magnitude is quantified by E_{learn} .

3 Part II: Research Directions

3.1 Section 8: Direction 1 — Remasking Error Bound in the L&Z Framework

3.1.1 8.1 Problem Statement

The L&Z framework (Theorem 2.1) provides a two-term decomposition of the total error into E_{learn} (fixed by training) and E_{fact} (controllable at inference time). The EB-Sampler (Theorem 4.2) is optimal for E_{fact} among all unmasking-only schedules. But the EB-Sampler commits each token exactly once and never revisits a committed token. In practice, tokens committed early (when context is sparse) may be committed incorrectly: by the time later tokens are revealed, the early commitment is inconsistent with the growing context. This is the **premature commitment** problem, and it is the failure mode that remasking strategies (ReMDM, confidence-based remasking) are designed to address.

The central open question for Direction 1 is:

Does adding a remasking step reduce E_{fact} below the EB-Sampler optimum?

The EB-Sampler is optimal for the *unmasking-only* problem. But remasking enlarges the search space: the algorithm is allowed to both unmask *and* remask. In this larger space, the EB-Sampler is no longer the optimal member. Direction 1 aims to prove a formal bound showing that adding remasking strictly reduces E_{fact} whenever premature commitments are present.

3.1.2 Target Theorem 8.1 (Remasking Reduces E_{fact}) [Conjecture]

Statement. Let EB_ε denote the EB-Sampler with entropy budget ε , and let $\text{EB}_\varepsilon + \text{remask}_\tau$ denote the combined algorithm that applies the EB-Sampler and additionally remasks all committed tokens j with confidence signal $\sigma_t^j = H(p_\theta(x^j | z_t)) > \tau$ at each step t . Then:

$$E_{\text{fact}}(\text{EB}_\varepsilon + \text{remask}_\tau) \leq E_{\text{fact}}(\text{EB}_\varepsilon) - \Delta(\tau, I, \varepsilon)$$

where $\Delta(\tau, I, \varepsilon) \geq 0$ and $\Delta > 0$ whenever there exists at least one position j with residual information gain $\delta_j(t) > 0$ (Definition 8.2 below) and $\sigma_t^j > \tau$.

3.1.3 8.2 Key Concepts

Definition 8.1 (Premature Commitment). Token j is a **premature commitment** at time t if it was first unmasked at step $t_j < t$ but its residual factorisation error — the information it would gain from being re-evaluated with the current context — is strictly positive.

Formally, define the **retrospective information gain**:

$$\delta_j(t) = H(p_\theta(x^j | z_{t_j})) - H(p_\theta(x^j | z_t))$$

where z_t is the current state (which has more context than z_{t_j} had at the time of commitment). $\delta_j(t) > 0$ means that, if we re-evaluated token j now, the model would be more confident (lower entropy) than it was when it originally committed j at step t_j .

Definition 8.2 (Tier 3 Confidence Signal). The **Tier 3 confidence signal** for committed token j at time t is:

$$\sigma_t^j = H(p_\theta(x^j \mid z_t))$$

the current model entropy at position j , evaluated by (conceptually) masking j and querying the model. This signal is computable at inference time without knowledge of x_0 .

The **remasking rule** with threshold τ : at each step t , for all currently committed tokens j , if $\sigma_t^j > \tau$, mask j back to $[M]$ and let the EB-Sampler re-unmask it in the next step.

3.1.4 8.3 Proof Strategy

The proof of Theorem 8.1 has three steps: decomposing E_{fact} into per-token commitment costs, showing that remasking reduces each such cost via the data processing inequality, and connecting the reduction to the threshold signal.

Step 1: Decompose E_{fact} into per-token commitment costs [Proof strategy]

We rewrite the total E_{fact} as a sum over individual token commitments:

$$E_{\text{fact}} = \sum_j E_{\text{fact}}(j, t_j)$$

where t_j is the step at which token j was first unmasked and $E_{\text{fact}}(j, t_j)$ is the factorisation error attributable to that commitment:

$$E_{\text{fact}}(j, t_j) = \mathbb{E} \left[\text{KL} \left(q(x^j \mid z_{t_j}, x_0) \parallel p_\theta(x^j \mid z_{t_j}) \right) \right]$$

This can be written as:

$$E_{\text{fact}}(j, t_j) = I^j(x) - I^j(x \mid \text{context at } t_j)$$

the information “lost” by not conditioning on the correlations between j and the tokens still masked at step t_j . When t_j is early (sparse context), this quantity is large.

Step 2: Remasking reduces the per-token commitment cost [Proved above, given the DPI]

Suppose we remask token j at time $t > t_j$ and re-commit at time t . The new commitment cost is:

$$E_{\text{fact}}(j, t) = I^j(x) - I^j(x \mid \text{context at } t)$$

By the data processing inequality (Lemma after Theorem 3.2): revealing more tokens (moving from z_{t_j} to z_t) can only decrease uncertainty. Formally:

$$I^j(x | z_t) \leq I^j(x | z_{t_j})$$

because z_t has more revealed tokens than z_{t_j} (we are further along in the reverse process), so conditioning on z_t provides more information.

Therefore:

$$E_{\text{fact}}(j, t) \leq E_{\text{fact}}(j, t_j)$$

and the gain from remasking is:

$$\Delta_j(t, t_j) = E_{\text{fact}}(j, t_j) - E_{\text{fact}}(j, t) = I^j(x | z_{t_j}) - I^j(x | z_t) = \delta_j(t) \geq 0$$

with $\Delta_j > 0$ whenever z_t provides strictly more information about x^j than z_{t_j} did.

Step 3: The threshold τ detects premature commitments [Proof strategy, partially conjectural]

We need to connect the observable signal $\sigma_t^j = H(p_\theta(x^j | z_t))$ to the unobservable quantity $\delta_j(t)$.

Claim: $\sigma_t^j > \tau$ implies $\delta_j(t) \geq \sigma_t^j - H(p_\theta(x^j | z_0))$.

Argument: At $t = 0$ (fully revealed context), the model entropy is $H(p_\theta(x^j | z_0)) \approx 0$ (the model is nearly certain about x^j when all context is available). Therefore:

$$\delta_j(t) = H(p_\theta(x^j | z_{t_j})) - H(p_\theta(x^j | z_t)) \approx H(p_\theta(x^j | z_{t_j})) - \sigma_t^j$$

and if $\sigma_t^j > \tau$, then:

$$\Delta_j \approx H(p_\theta(x^j | z_{t_j})) - H(p_\theta(x^j | z_t)) > H(p_\theta(x^j | z_{t_j})) - \sigma_t^j$$

which is positive as long as the model was not perfectly confident at the time of commitment.

Key gap and resolution approaches: The argument above uses $H(p_\theta(x^j | z_0)) \approx 0$ (model certainty at full context) and $H(p_\theta(x^j | z_{t_j})) \geq \sigma_t^j$ (entropy was at least as high at commitment as it is now). The second is guaranteed by the DPI: as more context is revealed, entropy can only decrease. The first requires connecting the model entropy to the true information $I^j(x | z_t)$:

(a) Consistency approach: Under perfect model assumption ($p_\theta = q$):

$$H(p_\theta(x^j | z_t)) = H(x^j | z_t) = I^j(x | z_t)$$

so the signal exactly equals the true information, and the connection is immediate.

(b) Imperfect model bound: For imperfect models, by the data processing inequality for entropy under channel perturbation:

$$|H(p_\theta(x^j | z_t)) - H(x^j | z_t)| \leq f(E_{\text{learn}}(t))$$

where f is an increasing function of the per-step learning error. This bounds the signal quality in terms of the training objective.

3.1.5 8.4 Attack Ideas

1. Information monotonicity approach. The data processing inequality is already in place (Step 2 above). The remaining work is to formalise the connection in Step 3. One approach: treat $H(p_\theta(x^j | z_t))$ as a surrogate for $I^j(x | z_t)$ and bound the surrogate error via E_{learn} . The resulting theorem would be: $\Delta(\tau, I, \varepsilon) \geq \tau - f(E_{\text{learn}})$, so remasking is guaranteed to reduce E_{fact} as long as the threshold exceeds the learning error.

2. Coupling approach. Couple the trajectories of EB-only and EB+remask. Define the coupling by running both algorithms on the same sequence of random bits, differing only in whether remasking steps are taken. Show that after any remasking step, the EB+remask trajectory is stochastically closer to π in the order induced by the information profile: its effective Σ^2 is smaller. By Theorem 3.1, lower Σ^2 implies lower E_{fact} — giving the reduction Δ .

3. Marginal distribution approach. After remasking token j at time t , the marginal distribution of x^j in the generated sequence becomes $p_\theta(x^j | z_t)$ instead of $p_\theta(x^j | z_{t_j})$. By KL monotonicity under conditioning:

$$\text{KL}(q(x^j | x_0) \| p_\theta(x^j | z_t)) \leq \text{KL}(q(x^j | x_0) \| p_\theta(x^j | z_{t_j}))$$

Each remasking step brings the marginal posterior closer to the true posterior. Summing over all remasked positions gives the total reduction Δ .

4. Equivalent Riemann reformulation. View the EB+remask algorithm as generating an effective information curve $I'(\alpha)$ that is smoother than the original $I(\alpha)$ (remasking redistributes committed information to later, better-contextualised steps). Show that remasking reduces the irregularity: $\text{Var}[I'] \leq \text{Var}[I] - \Delta$. By Theorem 3.1, this translates directly to $E_{\text{fact}}(\text{EB} + \text{remask}) \leq E_{\text{fact}}(\text{EB}) - C \cdot \Delta/T$.

3.1.6 8.5 Connection to Existing Results

Direction 1 directly extends the L&Z framework by adding the remasking case. The table of contributions would be:

Contribution	Source	Type
$\text{KL} \leq E_{\text{learn}} + E_{\text{fact}}$	L&Z 2025	Decomposition
E_{fact} minimised by easy-first unmasking	L&Z, EB-Sampler	Optimality
Correctors reduce KL exponentially in R	Informed Correctors 2024	Mixing

Contribution	Source	Type
Remasking reduces E_{fact} below EB-only	Thesis (Direction 1)	New bound

This is the most direct contribution: it takes the advisor’s framework and extends it to cover the one inference-time operation (remasking) that the original L&Z paper did not analyse.

3.2 Section 9: Direction 2 — Bridging the Riemann and MCMC Frameworks

3.2.1 9.1 Problem Statement

Two independent theoretical frameworks both conclude that uncertainty-guided strategies outperform uniform strategies. The L&Z Riemann thread (Section 3) says the advantage scales with $\Sigma^2 = \text{Var}_i[I^i(x)]$. The Informed Correctors MCMC thread (Section 6) says the advantage scales with $\text{Var}_d[s_d(z)]$ where $s_d = H(p_\theta(x^d \mid z_{\neq d}))$. The question is whether these two quantities are formally related. If $\Sigma^2 \propto \mathbb{E}[\text{Var}_d[s_d]]$, then the two frameworks are saying the same thing from different mathematical perspectives, and their results can be unified into a single theorem.

3.2.2 Target Theorem 9.1 (Framework Equivalence) [Open]

Statement. For a well-trained denoiser p_θ on data distribution π , and in the limit $t \rightarrow 0$ (vanishing masking):

$$\mathbb{E}_{x \sim \pi}[\Sigma^2(x)] = \mathbb{E}_{x \sim \pi}[\text{Var}_d[s_d(x)]]$$

where $s_d(x) = H(p_\theta(x^d \mid x_{\neq d}))$ is the surprise at position d for the fully revealed sequence x .

More generally, for $t > 0$: $\mathbb{E}_{z_t \sim q_t}[\text{Var}_d[s_d(z_t)]]$ is monotone in $\mathbb{E}[\Sigma^2]$, and the two framework predictions about strategy rankings agree.

3.2.3 9.2 The Two Quantities, Stated Precisely

Σ^2 (Riemann thread):

$$\Sigma^2(x) = \text{Var}_i[I^i(x)] = \frac{1}{L} \sum_{i=1}^L (I^i(x) - \bar{I}(x))^2$$

where $I^i(x) = H(x^i \mid x_{\neq i})$ and $\bar{I}(x) = \frac{1}{L} \sum_i I^i(x)$.

$\text{Var}[s_d]$ (MCMC thread):

At state z_t : $s_d(z_t) = H(p_\theta(x^d \mid z_{t,\neq d}))$, and:

$$\text{Var}_d[s_d(z_t)] = \frac{1}{L} \sum_{d=1}^L (s_d(z_t) - \bar{s}(z_t))^2$$

where $\bar{s}(z_t) = \frac{1}{L} \sum_d s_d(z_t)$.

At $t = 0$ (perfect model, fully revealed context):

$$s_d(x) = H(p_\theta(x^d \mid x_{\neq d})) = H(x^d \mid x_{\neq d}) = I^d(x)$$

where the second equality uses the perfect-model assumption. Therefore:

$$\text{Var}_d[s_d(x)] = \text{Var}_d[I^d(x)] = \Sigma^2(x)$$

The equivalence is exact at $t = 0$. The open question is whether it extends to $t > 0$.

3.2.4 9.3 Proof Strategy

Step 1: Exact equality at $t = 0$ [Proved above]

Under the perfect-model assumption, at $t = 0$:

$$s_d(x) = I^d(x) \implies \text{Var}_d[s_d(x)] = \Sigma^2(x)$$

This is exact and immediate. Taking expectations over $x \sim \pi$: $\mathbb{E}[\text{Var}_d[s_d]] = \mathbb{E}[\Sigma^2]$.

Step 2: Approximate equality for small t (Taylor expansion) [Proof strategy]

For small masking ratio $\alpha_t \ll 1$, each masked position i sees its token replaced by $[\mathbf{M}]$ with probability α_t and left unchanged with probability $1 - \alpha_t$. We can Taylor-expand the model entropy around the fully-revealed state x :

$$H(p_\theta(x^d \mid z_{t,\neq d})) = H(p_\theta(x^d \mid x_{\neq d})) + \alpha_t \cdot \left. \frac{\partial}{\partial \alpha} H(p_\theta(x^d \mid z_\alpha^{\neq d})) \right|_{\alpha=0} + O(\alpha_t^2)$$

The first-order correction $\frac{\partial}{\partial \alpha}[\cdot]$ measures how much the entropy at position d changes when we randomly mask each other position independently with probability α_t . This is a sum of per-position corrections:

$$\frac{\partial}{\partial \alpha} H(p_\theta(x^d \mid z_\alpha^{\neq d})) = \sum_{i \neq d} \left(H(p_\theta(x^d \mid z_\alpha^{\neq d}, \text{mask } i)) - H(p_\theta(x^d \mid z_\alpha^{\neq d})) \right)$$

Each term measures the increase in entropy at d when we additionally mask position i — which is exactly $I(x^d; x^i \mid z_\alpha^{\neq d, i})$, a pairwise mutual information. At $\alpha = 0$:

$$\left. \frac{\partial}{\partial \alpha} H(p_\theta(x^d \mid z_\alpha^{\neq d})) \right|_{\alpha=0} = \sum_{i \neq d} I(x^d; x^i \mid x_{\neq d, i})$$

This is the total “second-order information” at position d — how much masking other positions affects the uncertainty at d . Computing the variance of this across positions d would give the first-order correction to Σ^2 . The claim (requiring verification) is that this correction preserves

the ordering: $\text{Var}_d[s_d(z_t)] \propto \Sigma^2(x) + O(\alpha_t)$, with the proportionality constant and correction term expressible in terms of higher-order information quantities of π .

Step 3: Monotone relationship at general t [Conjecture]

For general t , both $\text{Var}_d[s_d(z_t)]$ and $\mathbb{E}[\Sigma^2]$ decrease as t increases (as more tokens are masked, positions become more uncertain and the profile flattens). The conjecture is that they decrease at the same rate — i.e., the ratio $\text{Var}_d[s_d(z_t)]/\mathbb{E}[\Sigma^2]$ remains bounded and bounded away from zero uniformly in t . If this holds, both quantities give the same ranking of strategies at all noise levels.

3.2.5 9.4 Attack Ideas

1. Taylor expansion approach. Implement Step 2 above explicitly. Compute the first- and second-order corrections to $\text{Var}_d[s_d(z_t)]$ as a power series in α_t . If the leading terms match those of a corresponding expansion of $\Sigma^2(t)$ (the effective information variance at noise level t), the equivalence follows to leading order.

2. Optimal transport approach. Both Σ^2 and $\text{Var}[s_d]$ are measures of “spread” on the information profile, viewed as a probability distribution on $[0, \log |\mathcal{V}|]$. Equivalence in the spirit of Theorem 9.1 may follow from the fact that both are second moments of this distribution (one under π , one under the model). Tools from optimal transport (in particular, the relationship between Wasserstein distance and moment convergence) may formalise this.

3. Direct computation on a 2-token model. Consider $L = 2$, $\mathcal{V} = \{0, 1\}$, and a joint distribution $\pi(x^1, x^2)$ parametrised by correlation $\rho \in [-1, 1]$. Compute Σ^2 and $\text{Var}[s_d]$ analytically as functions of ρ . If they are equal (or proportional) for all ρ , this strongly suggests the general equivalence and provides a clean formula for the proportionality constant. If they differ, the $L = 2$ example isolates exactly what breaks the equivalence and guides a refined conjecture.

4. Counterexample approach. If the equivalence is false, construct a distribution where Σ^2 is large but $\text{Var}[s_d]$ is small. One candidate: a distribution where information is heterogeneous across positions (high Σ^2) but the denoiser, because of its architecture, smooths out its entropy predictions (low $\text{Var}[s_d]$). This would show that the two frameworks give different guidance in practice, and the open question would be resolved by a separation result rather than a unification.

3.2.6 9.5 Why This Direction Is Compelling

The L&Z framework was developed by the thesis advisor (Zanella, 2025). The Informed Correctors framework (Zhao et al., 2024) was developed independently, building on DFM (Gat et al., 2024). Both were written around the same time, do not cite each other on the unification point, and arrive at superficially similar conclusions through entirely different mathematics (Riemann approximation theory vs. spectral gap theory of Markov chains). A formal bridge between these two frameworks would be the first unification result in this space.

The value of Theorem 9.1, if proven, is two-fold. Theoretically: it shows that the “non-uniformity of uncertainty” is a fundamental quantity that governs strategy performance regardless of which mathematical framework one uses, elevating it to the status of a natural invariant of the problem. Practically: it would allow results proven in the Riemann framework (such as Direction 1’s remasking bound) to be immediately translated into spectral gap

terms, and vice versa. The thesis would go from “extension of one paper” to “unification of two bodies of work” — a qualitatively stronger contribution.

3.3 Section 10: Direction 3 — EB-Sampler + Remasking: The Combined Algorithm

3.3.1 10.1 Problem Statement

The EB-Sampler is optimal for unmasking only. Direction 1 shows remasking reduces E_{fact} . But the combined EB+remask algorithm is not simply the sum of its parts: the unmasking and remasking steps interact.

When remasking returns a token to $[M]$, it changes the state z_t seen by the EB-Sampler at the next step. This new state was not produced by the standard forward process — it is a “targeted” masked state where specific high-uncertainty tokens are masked while others remain revealed. Training states have uniformly random masking patterns; the targeted masked state is out-of-distribution (OOD) for the model. This creates an OOD penalty that must be weighed against the benefit of the remasking.

The central question of Direction 3 is: **what is the optimal remasking threshold τ^* ?** Too aggressive (low τ): many tokens are remasked, creating OOD states and increasing E_{learn} ; the OOD penalty dominates and performance degrades. Too conservative (high τ): few tokens are remasked, premature commitments persist and E_{fact} remains high. Direction 3 aims to formalise this tradeoff and characterise τ^* .

3.3.2 Definition 10.1 (EB+Remask Algorithm)

At each step t (iterating from T down to 0): 1. **Sort:** For each masked position i , compute $H_i = H(p_\theta(x^i | z_t))$. Sort in ascending order. 2. **Unmask:** Greedily select positions S_t in ascending order of H_i until $\sum_{i \in S_t} H_i \leq \varepsilon$. Sample $x^i \sim p_\theta(x^i | z_t)$ for $i \in S_t$. 3. **Remask:** For all committed positions $j \notin S_t$ (previously unmasked), if $H(p_\theta(x^j | z_t)) > \tau$, set $z_t^j \leftarrow [M]$. 4. Repeat until all positions are committed.

The order of Steps 2 and 3 matters: we first unmask new easy tokens (bringing in more context), then evaluate committed tokens against the improved context. This ordering ensures the remasking step has access to the most current context.

3.3.3 Target Theorem 10.1 (Combined Algorithm Bound) [Conjecture]

Statement. The combined EB+remask algorithm satisfies:

$$E_{\text{total}}(\text{EB} + \text{remask}) \leq E_{\text{learn}} + E_{\text{fact}}(\text{EB}) - \Delta_{\text{remask}}(\tau, I) + E_{\text{interaction}}(\varepsilon, \tau)$$

where: - $\Delta_{\text{remask}}(\tau, I) \geq 0$ is the reduction in E_{fact} from remasking (established in Direction 1) - $E_{\text{interaction}}(\varepsilon, \tau)$ captures the interaction between unmasking and remasking - $E_{\text{interaction}}(\varepsilon, \tau) \leq C \cdot \varepsilon \cdot \tau$ for a constant C , which is small when both the entropy budget ε and the remasking threshold τ are small

3.3.4 10.2 The Interaction Term: Synergy and OOD Penalty

The interaction between unmasking and remasking has two components with opposite signs.

Positive interaction (synergy). After remasking token j at time t , the model’s context for subsequent EB-Sampler steps is updated: position j is now masked again, which changes the entropy estimates H_i for neighboring positions i . In many cases, this improves the sorting: positions that were incorrectly sorted before (because j ’s committed value was wrong) are now sorted correctly. This makes the subsequent EB steps more effective, reducing E_{fact} on later steps. The synergy gain is proportional to the correlation between token j and its neighbors — high when j was a “hub” token with many strong dependencies.

Negative interaction (OOD risk). The remasked state z'_t (with token j re-masked) has masking pattern that was not seen during training. Training masks tokens uniformly at random; the remasked state has a specific non-uniform pattern (only high-uncertainty tokens are masked). The model p_θ was not trained on such states, so its conditional distributions $p_\theta(x^i | z'_t)$ may be inaccurate for the remasked state. This increases the effective E_{learn} at the remasked positions. The OOD penalty is proportional to the fraction of tokens remasked and the model’s sensitivity to masking pattern non-uniformity.

3.3.5 10.3 Proof Strategy

Step 1: Decompose the total error by time step [Proof strategy]

$$E_{\text{total}} = \sum_t [E_{\text{fact,unmask}}(t) + E_{\text{fact,remask}}(t) + E_{\text{interact}}(t)]$$

The first term is the EB-Sampler contribution at step t (bounded by Theorem 4.1). The second term is the factorisation error reduction from remasking (bounded by Direction 1). The third term $E_{\text{interact}}(t)$ is the cross-term: does the remasking at step t affect the unmasking at step $t + 1$ and vice versa?

Step 2: Bound $E_{\text{interact}}(t)$ using the entropy budget [Proof strategy]

After remasking at step t , the effective information profile for step $t + 1$ is updated: let $\Sigma^2(t + 1 | \text{remask})$ denote the profile variance after remasking at step t . Define the synergy gain:

$$\delta\Sigma^2(t) = \Sigma^2(t + 1 | \text{no remask}) - \Sigma^2(t + 1 | \text{remask}) \geq 0$$

Remasking makes the profile smoother (more tokens are uncertain $\rightarrow$ profile is more uniform $\rightarrow$ Σ^2 decreases). By Theorem 3.1, lower Σ^2 implies lower $E_{\text{fact,unmask}}$ at step $t + 1$ by $C \cdot \delta\Sigma^2(t)/T$ per step. This is the synergy term.

Step 3: Bound the OOD penalty [Proof strategy]

Define the remasking fraction at step t : $f_t = |\{j : \sigma_t^j > \tau\}|/L$ (the fraction of committed tokens remasked at step t). The remasked state z'_t differs from any training-distribution state by having f_t targeted masked positions. The OOD gap is:

$$\text{KL}(q'_t \| q_t) \leq C \cdot f_t$$

where q'_t is the marginal distribution of the remasked state and q_t is the training-time marginal at masking level α_t . This uses the fact that each remasked token is an independent perturbation of the masking pattern, and each contributes independently to the KL. The resulting increase in E_{learn} is:

$$\Delta E_{\text{learn}}(\text{remask}) \leq C \cdot f_t \cdot E_{\text{learn}}(t)$$

where $E_{\text{learn}}(t)$ is the per-step learning error — the model’s sensitivity to the exact masking pattern is bounded by its overall inaccuracy.

Step 4: Combine into the interaction bound [Conjecture]

$$E_{\text{interact}}(t) = -C \cdot \delta \Sigma^2(t)/T + C' \cdot f_t \cdot E_{\text{learn}}(t)$$

$$E_{\text{interaction}}(\varepsilon, \tau) = \sum_t E_{\text{interact}}(t) \leq C \cdot \varepsilon \cdot \tau$$

The bound $\varepsilon \cdot \tau$ comes from: ε bounds the entropy budget (how much context changes at each unmasking step), and τ bounds the remasking threshold (only tokens with entropy above τ are remasked). When both are small, the algorithm is conservative and the interaction term vanishes. The joint smallness condition ensures that neither operation is too aggressive.

3.3.6 10.4 Attack Ideas

1. Monotone coupling. Couple EB-only and EB+remask trajectories by running both on the same random bits, differing only in the remasking step. After any remasking step, define the “lead” of the EB+remask trajectory as a measure of how much it has reduced Σ^2 relative to EB-only. Show that the lead is non-decreasing in t (each remasking step either maintains or increases the lead). The interaction term is then bounded by the worst-case decrease in lead from a single OOD state.

2. Fixed-point analysis. The combined algorithm has a natural fixed point: a state where no remasking is triggered ($\sigma_t^j \leq \tau$ for all committed j). Near this fixed point, the remasking fraction f_t is small, and the interaction term is second-order in $(\tau - \tau^*)$ where τ^* is the critical threshold at which the first remasking event occurs. A Taylor expansion around the fixed point gives $E_{\text{interaction}} = O((\tau - \tau^*)^2)$, which is small near the optimal threshold.

3. Calculus of variations for optimal τ . Treat the remasking fraction as a function of time: $f : [0, T] \rightarrow [0, 1]$, $\$f(t) = \$$ fraction of tokens remasked at step t . The total error is a functional:

$$E_{\text{total}}[f] = E_{\text{fact}}(\text{EB}) - \int_0^T g(f(t)) dt + \int_0^T h(f(t)) \cdot E_{\text{learn}}(t) dt$$

where $g(f)$ is the synergy gain (concave, increasing in f) and $h(f) \cdot E_{\text{learn}}(t)$ is the OOD penalty (convex, increasing in f). Minimising over $f(t)$ by Euler-Lagrange equations gives the optimal schedule $f^*(t)$, which translates to an optimal threshold $\tau^*(t)$ as a function of

time. The prediction: $\tau^*(t)$ decreases as t increases (be more selective about remasking late in the process when context is rich and OOD risk is highest).

4. Empirical validation. Using the existing MDLM-OWT experimental infrastructure, run a sweep over $\tau \in \{0.1, 0.2, \dots, 0.9\}$ (in bits) for a fixed ε . The MAUVE score as a function of τ will display a clear maximum at τ^* . The theoretical prediction from the calculus of variations (Attack 3) gives a closed-form for τ^* in terms of E_{learn} and the information profile. If the theoretical τ^* matches the empirical optimum, this simultaneously validates the theory and provides a practical algorithm selection tool.

3.3.7 10.5 Connection to the Empirical Results

The step-sweep results from the current experimental infrastructure directly instantiate the tradeoff analysed by Direction 3:

- **remdm-conf** (confidence-based remasking, effectively low τ): aggressively remasks any token below a confidence threshold. At $T = 1000$ steps, MAUVE collapses to 0.325 (below its $T = 128$ value of 0.440). This is consistent with the OOD penalty dominating: aggressive remasking creates highly targeted masked states that are far from the training distribution, increasing E_{learn} catastrophically at high step counts.
- **remdm-loop** (loop-based remasking, effectively high τ): applies remasking more conservatively, only reconsidering tokens when the full context has changed substantially. MAUVE improves monotonically from 0.396 ($T = 128$) to 0.684 ($T = 1000$). This is consistent with the synergy term dominating: conservative remasking stays near the training distribution and progressively reduces E_{fact} without triggering the OOD penalty.
- **mdlm** (no remasking): pure EB-Sampler baseline. MAUVE peaks at $T = 256$ (0.740) and declines at higher step counts. This “diversity window” — a peak followed by decline — is explained by the EB-Sampler behaviour: at $T = 256$, the step size is small enough that factorisation error is manageable, but at $T = 1000$, the very small steps introduce other artefacts (the model is queried many times on very similar near-complete states, which can concentrate the distribution excessively).

Direction 3’s theorem would formalise this observation: conf collapses because it remasks too aggressively (remasking fraction f_t large, OOD penalty $C \cdot f_t \cdot E_{\text{learn}}$ dominates), while loop succeeds because it remasks conservatively (remasking fraction f_t small, synergy term dominates). The formal theorem would give an explicit expression for the crossover threshold τ^* where the strategy changes from beneficial to harmful.

4 Open Problems Summary

Problem	Status	Direction
Remasking reduces E_{fact} below EB-only	Conjectured	Dir 1
Tier 3 signal σ_t^j sufficient for E_{fact} reduction	Conjectured	Dir 1

Problem	Status	Direction
$\Sigma^2 = \text{Var}[s_d]$ formally	Open	Dir 2
Optimal remasking schedule characterisable in closed form	Open	Dir 2/3
$E_{\text{interaction}}(\varepsilon, \tau) \leq C \cdot \varepsilon \cdot \tau$	Conjectured	Dir 3
Optimal τ^* expressible in terms of I and E_{learn}	Open	Dir 3

The three directions are nested from most concrete to most ambitious. **Direction 1** is the core theorem: extending the L&Z decomposition to the remasking case. The key tools — the data processing inequality, the KL triangle inequality, and the Riemann error decomposition — are already in hand (developed in Part I), and the main gap (connecting the model entropy signal to the true information) is a well-defined problem with clear solution paths. This is the guaranteed deliverable, most directly building on the advisor’s framework.

Direction 2 is the ambitious stretch: bridging the Riemann and MCMC frameworks into a single unification theorem. The $t = 0$ base case is already proved (it is an immediate consequence of the perfect-model assumption), and the path to the general case is laid out via Taylor expansion and monotonicity. A successful Direction 2 result would be the strongest possible contribution, showing that the thesis goes beyond extending one paper to synthesising two independent lines of work.

Direction 3 is the experimental backbone: it formalises the interaction between unmasking and remasking, providing both a rigorous analysis of the combined algorithm and a direct theoretical explanation for the observed empirical results. The threshold sweep (Attack 4 above) is immediately implementable with existing infrastructure. Direction 3 validation feeds back into Direction 1 by providing empirical estimates of Δ_{remask} and the OOD penalty, which can guide the choice of tools for the formal proof. The three directions can be pursued in parallel, with Direction 3 empirical work providing ground truth that disciplines the theoretical development of Directions 1 and 2.